Age and systemic acid-base equilibrium: analysis of published data.

To investigate whether systemic acid-base equilibrium changes with aging in normal adult humans, we reviewed published articles reporting the acid-base composition of arterial, arterialized venous, or capillary blood in age-identified healthy subjects. We extracted or calculated blood hydrogen ion concentration ($[H^+]$), plasma bicarbonate concentration ($[HCO_3^{(-)}]$), blood PCO_2, and age, and computed a total of 61 age-group means, distributed among eight 10-year intervals from age 20 to 100 years. Using linear regression analysis, we found that with increasing age, there is a significant increase in the steady-state blood $[H^+]$ ($p < .001$), and reduction in steady-state plasma $[HCO_3^{(-)}]$ ($p < .001$), indicative of a progressively worsening low-level metabolic acidosis. Blood PCO_2 decreased with age ($p < .05$), in keeping with the expected respiratory adaptation to metabolic acidosis. Such age-related increasing metabolic acidosis may reflect in part the normal decline of renal function with increasing age. The role of age-related metabolic acidosis in the pathogenesis of the degenerative diseases of aging warrants consideration.